

I review the empirical pathologies of deep reinforcement learning, their causes, and the diagnostic tools that expose them.

0.1 The Moving Target Problem

In supervised learning, the loss function is a *fixed function* of the training data and model parameters. Deep reinforcement learning does not enjoy this property. Each gradient step moves both the current value estimates and the targets simultaneously, creating a “nonstationary” optimization landscape. The target network heuristic introduced by Mnih et al. (2015) slows target drift by periodically freezing a copy of the network, but does not eliminate it. Therefore Bellman residual is a poor proxy for the accuracy of the value function.

Fujimoto et al. (2022) formalize this observation. Let Q^π denote the true value function for policy π , let $\Delta(s, a) = Q(s, a) - Q^\pi(s, a)$ denote the value error, and let $\varepsilon(s, a) = Q(s, a) - (r + \gamma \mathbb{E}_{s', a'}[Q(s', a')])$ denote the Bellman error. Substituting the definition of Q^π yields the key identity: $\varepsilon(s, a) = \Delta(s, a) - \gamma \mathbb{E}_{s', a'}[\Delta(s', a')]$. The Bellman error is a *difference* of value errors at consecutive states, not the value error itself. If the errors $\Delta(s, a)$ and $\Delta(s', a')$ are correlated across time—the network is wrong in the same direction at successive state-action pairs—they cancel in the difference and the Bellman error is small regardless of how large the individual errors are.¹

The second failure mode is specific to finite datasets: Fujimoto et al. (2022, Corollary 1) show that over an incomplete dataset, zero Bellman error is consistent with arbitrarily large value error, because the network can fit unobserved successor pairs to whatever values make the observed residuals vanish.²

The dual failure mode appears in policy gradient methods: Ilyas et al. (2020) find that even when PPO’s surrogate objective improves monotonically, episode return can plateau or decline, because the surrogate gradient is poorly aligned with the gradient of the true return.³

¹In the extreme case, shifting all Q-values by the constant $c/(1 - \gamma)$ leaves the Bellman error unchanged at zero while increasing value error by $c/(1 - \gamma)$.

²The Bellman equation uniquely identifies Q^π when enforced over the entire MDP, but over an incomplete dataset it admits infinitely many solutions. Whenever a transition (s', a') that is reachable from the dataset is not itself in the dataset, the network is free to set $Q(s', a')$ at unobserved pairs to whatever value makes the residual vanish on the observed ones, unconstrained by any loss. A network can thus reach near-zero training loss while the value function remains arbitrarily inaccurate over the full state space.

³Ilyas et al. (2020) examine the surrogate objective used in Proximal Policy Optimization (Schulman et al., 2017). The PPO clipping mechanism is designed to keep policy updates within a trust region by bounding the

0.2 The Reproducibility Crisis and Sensitivity to Random Seeds

Henderson et al. (2018) trained five leading policy gradient algorithms (PPO, TRPO, DDPG, TD3, SAC) on six MuJoCo benchmark environments, holding all hyperparameters fixed and varying only the random seed. The resulting learning curves from different seeds were non-overlapping: a seed that performed well under one algorithm performed comparably to a different algorithm’s best seeds, making cross-algorithm comparison unreliable.⁴ Agarwal et al. (2021) quantify the damage: comparing point estimates from 5 runs per task on Atari 100k yields Type I error exceeding 50%, meaning a random noise injection appears beneficial in half of all comparisons.

Agarwal et al. (2021) propose the *interquartile mean* (IQM) as a replacement for mean and median when comparing algorithms. The IQM discards the top and bottom 25% of runs before averaging, reducing sensitivity to outlier seeds. Using these tools and stratified bootstrap confidence intervals, Agarwal et al. (2021) find that several widely-cited algorithmic improvements on Atari 100k vanish or reverse when statistical uncertainty is accounted for.⁵⁶

0.3 Value Overestimation and Spikes

Q-learning uses the Bellman optimality update

$$Q(s, a) \leftarrow r + \gamma \max_{a'} Q(s', a'), \quad (1)$$

where the maximum is taken over the estimated Q-values of all actions at the successor state. Thrun and Schwartz (1993) identify a positive bias intrinsic to this update: if the Q-value estimates contain noise with mean zero, the maximum over noisy estimates is biased upward by Jensen’s inequality. An agent that uses a single network for both action

probability ratio $\pi_{\theta}(a|s)/\pi_{\theta_{\text{old}}}(a|s)$. The gradient of the surrogate is poorly aligned with the gradient of the true return, particularly in later training phases where the policy has diverged from the behavior policy used to collect the replay data. The loss metric that practitioners monitor throughout training is measuring something other than what they care about.

⁴Differences as large as 2,000 points in final episode return arose from seed variation alone.

⁵They also introduce performance profiles, which plot the fraction of tasks and seeds where an algorithm achieves performance above a threshold τ , as τ varies from 0 to the maximum. Performance profiles reveal the full shape of the score distribution rather than collapsing it to a single statistic.

⁶The fragility extends to hyperparameters. Eimer et al. (2023) conduct a systematic study of hyperparameter sensitivity across 6 algorithms and 17 environments, finding that default hyperparameters from published papers perform competitively in the specific environments used in those papers but generalize poorly across environments. Patterson et al. (2024) synthesize these findings into an empirical design handbook, recommending at minimum 10 seeds per configuration, IQM-based comparisons, and preregistration of hyperparameter search protocols.

selection ($\arg \max$) and value estimation ($\max Q$) systematically overestimates the values of every state it visits, biasing the Bellman target upward at every update step. The bias compounds through bootstrapping: overestimated targets produce overestimated updates, which produce further overestimated targets.

van Hasselt (2010) propose double Q-learning as a remedy: maintain two independent Q-networks Q_A and Q_B . Use Q_A to select the greedy action at s' , but use Q_B to evaluate that action. Because the two networks are trained on different data, their errors are approximately independent, and the positive bias largely cancels. van Hasselt et al. (2016b) implement this as Double DQN, using the online network for action selection and the periodically-frozen target network for evaluation. On 49 Atari games, Double DQN reduces overestimation by a factor of 3–5 and improves median performance by 20% relative to DQN.

Fujimoto et al. (2018) observe that Double DQN’s correction is incomplete in continuous-action settings, where the target network and online network remain correlated through shared updates. They propose Clipped Double Q-learning: compute two Q-value estimates Q_1, Q_2 with separate networks trained on the same data, and use $y = r + \gamma \min(Q_1(s', a'), Q_2(s', a'))$ as the Bellman target. The minimum operator introduces pessimistic underestimation, which is conservative but avoids the explosive positive bias.⁷

DQN prevents outright divergence, but van Hasselt et al. (2018) find that *soft divergence*, defined as a temporary spike in value estimates by more than 10% followed by recovery, occurs in the majority of DQN training runs across 57 Atari games.⁸ The deadly triad does not announce itself as a training failure: the value estimates may spike and recover, leaving no visible trace in the loss curve while corrupting the policy.

Kumar et al. (2021) describe a continuous manifestation of the deadly triad: *implicit under-parameterization*.⁹

⁷Ciosek et al. (2019) note that clipped double Q can cause excessive pessimism under high uncertainty, proposing optimistic actor-critic as a counterweight.

⁸Larger networks diverge more frequently than smaller ones, counter to the usual intuition that more expressive models should generalize better.

⁹Neural networks trained with bootstrapped TD objectives progressively lose their effective rank, with fewer and fewer neurons contributing distinct directions in the representation. This rank collapse is silent (training loss continues to decrease) but the network’s ability to represent new information degrades over time, a form of capacity loss distinct from but related to plasticity loss.

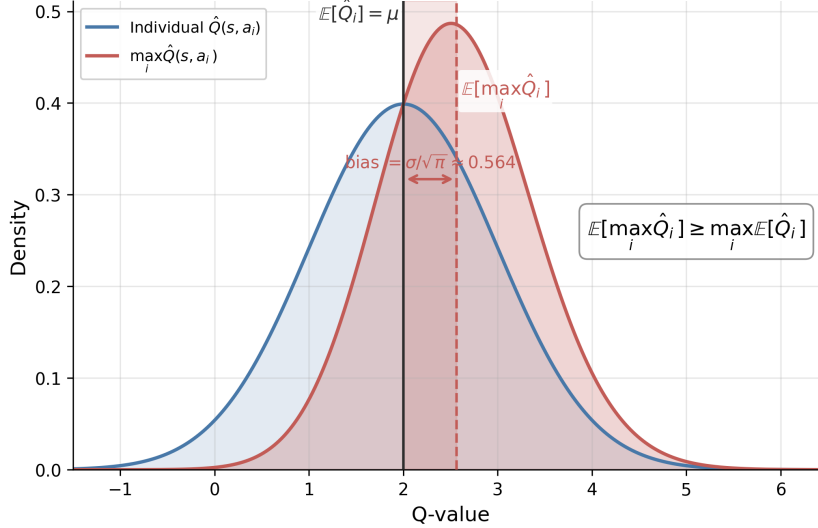


Figure 1: Jensen-inequality illustration of overestimation bias with $n = 2$ iid Gaussian Q-estimates. Blue: density of individual $\hat{Q}(s, a_i) \sim \mathcal{N}(\mu, \sigma^2)$. Red: density of $\max_i \hat{Q}(s, a_i)$, shifted right by $\sigma/\sqrt{\pi} \approx 0.56$. The shaded gap is the bias. With $n = 100$ actions the bias reaches 2.51σ . Empirical DQN-vs-Double-DQN learning curves are reported in [van Hasselt \(2010\)](#); [van Hasselt et al. \(2016b\)](#) and not reproduced here.

0.4 Plasticity Loss and Primacy Bias

A network that trains well at step $t = 100$ may be incapable of learning at step $t = 100,000$, even if the data quality at the later step is higher. [Lyle et al. \(2022\)](#) call this *capacity loss*: the network’s ability to update its own weights degrades progressively during training, measured by the fraction of effective parameters and the network’s ability to fit new random labels. [Lyle et al. \(2023\)](#) extend this to *plasticity loss*, identifying dead ReLU neurons, weight norm growth, and feature rank collapse as three distinct mechanisms, not all of which co-occur.

[Nikishin et al. \(2022\)](#) identify *primacy bias* as a specific cause. Because the replay buffer is filled incrementally, early transitions are oversampled relative to later ones, and the network over-fits to early environment transitions, corrupting representations throughout the remainder of training.¹⁰

The proposed remedies fall into three categories: periodic resets, continual backprop-

¹⁰[Nikishin et al. \(2022\)](#) show that 100 initial “priming” steps of excessive gradient updates degrade a SAC agent’s performance for hundreds of thousands of subsequent steps. [Sokar et al. \(2023\)](#) measure the fraction of dormant neurons (units with near-zero activation across the replay buffer) accumulating monotonically during training. [Dohare et al. \(2024\)](#) find that standard deep networks lose all plasticity within a few million gradient steps in continual learning tasks.

agation, and architectural interventions.¹¹

0.5 Implementation Dominates Algorithmic Innovation

Engstrom et al. (2020) find that PPO with the clipping mechanism disabled performs indistinguishably from the full algorithm; TRPO (Schulman et al., 2015) with the same code-level additions matches PPO.¹² Andrychowicz et al. (2021) identify observation normalization, orthogonal initialization, and learning rate annealing as the three choices accounting for most variance across 250,000 agents and 250 hyperparameter configurations. Huang et al. (2022) catalog 37 implementation details required to reproduce PPO on Atari;¹³ omitting any produces materially different results.

The most consequential implementation detail is the distinction between termination and truncation (Pardo et al., 2018). In reinforcement learning, an episode can end for two distinct reasons: *termination*, where the environment reaches a natural absorbing state (the pole falls in CartPole, the robot falls in locomotion tasks), and *truncation*, where the episode is cut short by an external time limit. At a termination, the value of the successor state is zero: $V(s_{\text{term}}) = 0$. At a truncation, the episode is merely paused and the successor state has non-zero value: $V(s_{\text{trunc}}) \neq 0$. Treating truncated transitions as terminated substitutes zero for a non-zero bootstrap value at every time limit boundary, corrupting every Bellman update in the vicinity of episode boundaries. Pardo et al. (2018) show that this conflation degrades performance by 20–40% on standard MuJoCo benchmarks.¹⁴

0.6 Replay Buffer Pathologies and Reward Scaling

Experience replay (Lin, 1992) decouples the data collection and learning processes, allowing a single transition to be used for multiple gradient updates. The replay ratio (the

¹¹Periodic resets reinitialize the last layers of the network while retaining the replay buffer, allowing the agent to forget overfit representations without discarding experience (Nikishin et al., 2022; D’Oro et al., 2023). Continual backpropagation replaces neurons with near-zero utility at each gradient step rather than waiting for a global reset (Dohare et al., 2024). Architectural interventions—layer normalization (Lyle et al., 2025), orthogonal initialization, spectral normalization—reduce the rate at which plasticity is lost by stabilizing gradient magnitudes and preventing weight norm growth.

¹²The non-clipping components that suffice are: observation normalization, reward normalization, value function clipping, global gradient norm clipping, orthogonal weight initialization, and the Adam optimizer.

¹³These include reward clipping to $[-1, 1]$, frame stacking to 4 frames, a specific episode termination convention, and a numerically stable normalization of the advantage estimate.

¹⁴The Gymnasium API (Towers et al., 2024) enforces the distinction by returning separate `terminated` and `truncated` flags, but most pre-2022 codebases conflate them in the `done` flag.

number of gradient updates per environment step) governs the trade-off between sample efficiency and data staleness. Zhang and Sutton (2017) show that increasing the replay ratio beyond a modest threshold degrades performance.¹⁵

Schaul et al. (2016) propose prioritized experience replay (PER).¹⁶ Fedus et al. (2020) revisit PER on large-scale Atari experiments and find that uniform sampling from a large enough buffer matches or outperforms PER, while being simpler to implement and tune.

Reward scaling introduces a separate class of failure modes. Standard DQN (Mnih et al., 2015) clips rewards to $[-1, +1]$ across all environments to stabilize training. van Hasselt et al. (2016a) observe that reward clipping changes the objective: clipped rewards make all positive events equivalent regardless of magnitude, so the agent learns to maximize the frequency of positive events rather than their cumulative value. This substitution can produce policies that are locally rational under the clipped reward but qualitatively suboptimal under the true reward. van Hasselt et al. (2016a) propose PopArt as a remedy.¹⁷

Skalse et al. (2022) formalize reward hacking and show that any non-constant proxy reward can in principle be exploited by a sufficiently capable optimizer.¹⁸

0.7 Simulation Study: Bellman Error and Value Error in Offline Policy Evaluation

The MDP uses $s = (k, z)$ with k on a 50-point log-spaced capital grid and $z \in \{0.9, 1.1\}$ following a Markov chain with persistence 0.8. Actions are next-period capital choices on the same grid. Reward is $\log(zk^\alpha - k')$ with $\alpha = 0.36$, $\beta = 0.96$. Rewards are shifted by $-\bar{r}$ (mean reward over feasible pairs) to center Q^* near zero, which avoids initialization issues without altering the optimal policy. The offline dataset $\mathcal{D}$ consists of $T = 2,000$ transitions simulated from the value-iteration optimal policy on the discretized

¹⁵At high replay ratios, the distribution shift between the current policy and the behavior policy that generated the stored data grows large enough to violate the off-policy assumptions of Q-learning. The relationship is non-monotone and environment-dependent, making replay buffer size a sensitive hyperparameter with no universal default.

¹⁶PER samples transitions with probability proportional to the magnitude of their TD error, on the argument that high-error transitions are the most informative.

¹⁷PopArt normalizes targets to have unit variance while adjusting the output layer so that the policy remains invariant to the normalization. PopArt allows consistent learning across reward scales spanning several orders of magnitude without reward clipping.

¹⁸Skalse et al. (2022) define a proxy as *unhackable* if increasing expected proxy return cannot decrease expected true return. Their main result states that for the set of all stochastic policies, two reward functions are unhackable only if one of them is constant. For deterministic policies and finite policy sets, non-trivial unhackable pairs exist, but the conditions are stringent.

grid, which agrees with the closed-form rule $k^*(k, z) = \alpha\beta zk^\alpha$ on 94% of states (the residual disagreement is single-grid-step rounding at the bottom of the capital grid). Since the optimal policy concentrates capital near its steady state, $\mathcal{D}$ covers only 11 of the 4,795 feasible (s, a) pairs—0.2% coverage—the distribution mismatch condition in Fujimoto et al. (2022, Corollary 1).

Two algorithms are trained on $\mathcal{D}$.¹⁹ BRM minimizes $\mathbb{E}_{\mathcal{D}}[(Q(s, a) - (r + \gamma \max_{a'} Q(s', a')))^2]$ where both $Q(s, a)$ and $Q(s', a')$ use the current network; gradients flow through both sides simultaneously.²⁰ The key consequence is that the network can zero the residual at an observed pair (s, a) by co-moving $Q(s, a)$ and $Q(s', a')$ together, rather than moving either toward Q^* . This is the opposite of supervised learning, where labels are fixed external targets that do not move with the model weights. FQE prevents this by using a frozen target network for $Q(s', a')$; the network must reduce $Q(s, a)$ toward a target that does not respond to its own gradient steps. Every 500 steps we record the Bellman error on $\mathcal{D}$ (current network on both sides) and the value error $\frac{1}{|\mathcal{F}|} \sum_{(s,a) \in \mathcal{F}} |Q_\theta(s, a) - Q^*(s, a)|$ over all 4,795 feasible pairs. As a definitional reference panel, OLS regression of $\log c$ on $\log k$ and $\log z$ is estimated on expanding windows.²¹

The OLS panel records Pearson $r(\text{MSE}, R^2) = -1.000$ (Table 1); on a fixed test set with fixed SS_{tot} , $R^2 = 1 - n_{\text{test}} \cdot \text{MSE} / SS_{\text{tot}}$ is an exact affine transform of MSE, so the Pearson correlation is -1 by construction rather than an empirical finding. The panel is plotted as an algebraic reference, a setting in which the training criterion and the held-out metric are pinned to each other, which is what supervised learning on a fixed test set delivers. The RL panel does not have this property. Across 10 seeds BRM drives the Bellman residual on $\mathcal{D}$ to a mean of 1.4×10^{-4} , roughly $1,500\times$ lower than FQE’s mean of 0.209 (median ratio $1,846\times$; three of ten BRM seeds drove BE below 5×10^{-5} , i.e. to numerical noise). Value error on the full MDP is statistically indistinguishable between the two methods: BRM mean VE = 7.380, FQE mean VE = 7.310, Welch

¹⁹50,000 gradient steps per seed, 10 seeds; two-layer MLP with 64 hidden units, Adam at 5×10^{-4} ; target network updated every 500 steps.

²⁰The implementation uses the Bellman-optimality target $\max_{a'} Q(s', a')$ rather than the policy-evaluation target $Q(s', a' \sim \pi)$ of Fujimoto et al. (2022, Section 2); both algorithms are therefore FQI-flavored variants of the Fujimoto formulations. Because $\mathcal{D}$ is generated by the optimal policy π^* , the two targets coincide on the data manifold and diverge off-manifold (where value error is evaluated). The qualitative BE-vs-VE pattern reported below survives the substitution; the comparison concerns value estimation, not control. Feasibility is enforced at the target step by a -10^9 mask on infeasible (s', a') pairs, so the network is told the constraint set but not the optimal action within it.

²¹Noise $\sigma = 0.30$; windows from $n = 10$ to 2,000; held-out test set of 500 transitions.

$t = 0.57$ on $n = 10$ seeds each ($p = 0.58$), a difference under one percent of either mean. The BE-vs-VE comparison is one of value estimation rather than control; both methods produce Q -functions whose greedy policies agree with π^* on roughly 6–8% of states, so neither delivers a usable decision rule under 0.2% coverage. The Fujimoto et al. (2022) critique concerns value estimates rather than policies, and that critique reproduces here, since minimizing Bellman residual on $\mathcal{D}$ does not approximate Q^* off-support. Both mechanisms from Section 0.1 operate, namely error cancellation on the 11 observed pairs and unconstrained $Q(s', a')$ at the 4,784 unobserved pairs.

Method	Final BE on $\mathcal{D}$	Final VE (all states)	VE/BE ratio	Policy
BRM (mean $\pm$ SE)	$(1.39 \pm 0.42) \times 10^{-4}$	7.380 ± 0.064	$1,244,793 \pm 676,157$	7
BRM (median)	1.04×10^{-4}	7.364	72,306	7
FQE (mean $\pm$ SE)	0.2087 ± 0.0394	7.310 ± 0.107	44.4 ± 6.8	6
FQE (median)	0.1911	7.237	37.2	6
OLS ($n = 2,000$, fixed test set)	OOS MSE = 0.0959	OOS $R^2 = 0.112$	$r(\text{MSE}, R^2) = -1.000$	

Table 1: Bellman error on dataset $\mathcal{D}$ and value error on the full MDP for BRM and FQE trained on offline Brock–Mirman data, with $n = 10$ seeds per method. BE is mean squared Bellman error evaluated with the current network on both sides; VE is mean absolute deviation from Q^* over all 4,795 feasible state-action pairs; SE is the standard error of the mean across seeds. Policy agreement is the fraction of states where $\arg \max_a Q_\theta(s, a) = \pi^*(s)$. The OLS row reports a single fixed-test-set fit; on that test set $r(\text{MSE}, R^2) = -1$ is an algebraic identity (see prose).

0.8 Discussion and Recommendations

Track episode return and policy entropy alongside training loss; entropy collapse and stagnating return are early warning signs of plasticity loss (Section 0.4). Use PPO or SAC as default baselines before implementing custom algorithms. Report at least 10 seeds per configuration with IQM-based comparisons (Section 0.2).

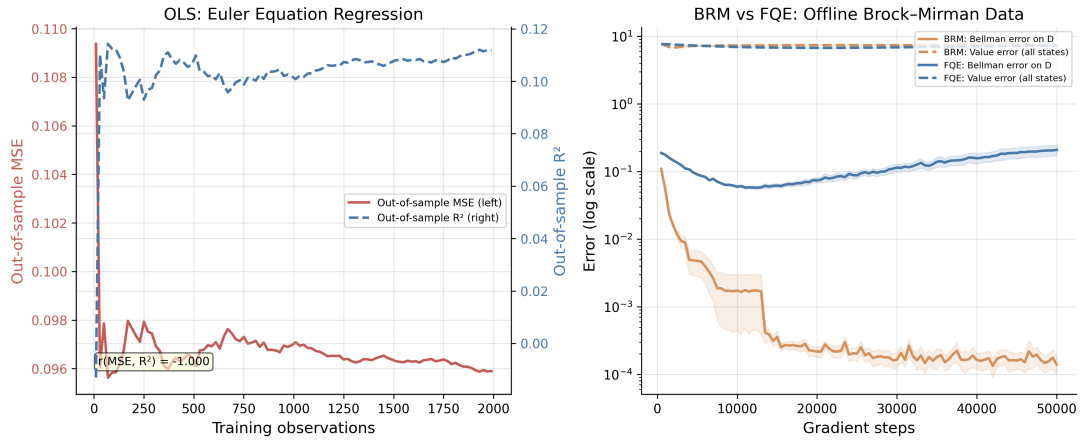


Figure 2: Left: OLS regression of log consumption on log capital and log productivity, estimated on expanding windows from the Brock–Mirman optimal policy. Out-of-sample MSE (left axis, red) and out-of-sample R^2 (right axis, blue) track each other with Pearson $r = -1.000$. Right: BRM (orange) and FQE (blue) trained on offline Brock–Mirman data $\mathcal{D}$ ($T = 2,000$ transitions, 0.2% state-space coverage); note the log scale on the y -axis. Solid lines show Bellman error on $\mathcal{D}$ (current network both sides); dashed lines show mean absolute value error against Q^* over all 4,795 feasible state-action pairs; shaded bands are ± 1 SE over 10 seeds.

References

- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- Marcin Andrychowicz, Anton Raichuk, Piotr Stańczyk, Manu Orsini, Sertan Girgin, Raphael Marinier, Leonard Hussenot, Matthieu Geist, Olivier Pietquin, Marcin Michalski, Sylvain Gelly, and Olivier Bachem. What matters for on-policy deep actor-critic methods? a large-scale study. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- Kamil Ciosek, Quan Vuong, Robert Lierowski, and Katja Hofmann. Better exploration with optimistic actor-critic. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Shibhansh Dohare, J. Fernando Hernandez-Garcia, Qingfeng Lan, Parash Rahman, A. Mahmoud, and Richard S. Sutton. Loss of plasticity in deep continual learning. *Nature*, 632:768–774, 2024.
- Pierluca D’Oro, Max Schwarzer, Evgenii Nikishin, Pierre-Luc Bacon, Marc G. Bellemare, and Aaron Courville. Sample-efficient reinforcement learning by breaking the replay ratio barrier. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.
- Theresa Eimer, Marius Lindauer, and Roberta Raileanu. Hyperparameters in reinforcement learning and how to tune them. In *Proceedings of the 40th International Conference on Machine Learning*, Proceedings of Machine Learning Research, pages 9104–9149, 2023.
- Logan Engstrom, Andrew Ilyas, Shibani Santurkar, Dimitris Tsipras, Firdaus Janoos, Larry Rudolph, and Aleksander Madry. Implementation matters in deep RL: A case study on PPO and TRPO. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020.
- William Fedus, Prajit Ramachandran, Rishabh Agarwal, Yoshua Bengio, Hugo Larochelle, Mark Rowland, and Marc G. Bellemare. Revisiting fundamentals of experi-

- ence replay. In *Proceedings of the 37th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 2020.
- Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *Proceedings of the 35th International Conference on Machine Learning*, Proceedings of Machine Learning Research, pages 1587–1596, 2018.
- Scott Fujimoto, David Meger, Doina Precup, Ofir Nachum, and Shixiang Shane Gu. Why should I trust you, Bellman? the Bellman error is a poor replacement for value error. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2022.
- Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.
- Shengyi Huang, Rousslan Fernand Julien Dossa, Antonin Raffin, Anssi Kanervisto, and Weng Wang. The 37 implementation details of proximal policy optimization. *ICLR Blog Track*, 2022.
- Andrew Ilyas, Logan Engstrom, Shibani Santurkar, Dimitris Tsipras, Firdaus Janoos, Larry Rudolph, and Aleksander Madry. A closer look at deep policy gradients. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020.
- Aviral Kumar, Rishabh Agarwal, Dibya Ghosh, and Sergey Levine. Implicit underparameterization inhibits data-efficient deep reinforcement learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- Long-Ji Lin. *Self-Improving Reactive Agents Based on Reinforcement Learning, Planning and Teaching*. PhD thesis, Carnegie Mellon University, 1992. Also published in *Machine Learning*, 8(3–4):293–321, 1992.
- Clare Lyle, Mark Rowland, and Will Dabney. Understanding and preventing capacity loss in reinforcement learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

- Clare Lyle, Mark Rowland, Will Dabney, Marta Kwiatkowska, and Yarin Gal. Understanding plasticity in neural networks. In *Proceedings of the 40th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 2023.
- Clare Lyle, Zeyu Zheng, Evgenii Nikishin, Bernardo Avila Pires, Razvan Pascanu, and Will Dabney. Disentangling causes of plasticity loss in neural networks. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dhharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.
- Evgenii Nikishin, Max Schwarzer, Pierluca D’Oro, Pierre-Luc Bacon, and Aaron Courville. The primacy bias in deep reinforcement learning. In *Proceedings of the 39th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 2022.
- Fabio Pardo, Arash Tavakoli, Vitaly Levдик, and Petar Kormushev. Time limits in reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 2018.
- Andrew Patterson, Samuel Neumann, Martha White, and Adam White. Empirical design in reinforcement learning. In *arXiv preprint arXiv:2304.01315*, 2024.
- Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2016.
- John Schulman, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz. Trust region policy optimization. *Proceedings of the 32nd International Conference on Machine Learning*, pages 1889–1897, 2015.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

- Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Ghada Sokar, Rishabh Agarwal, Pablo Samuel Castro, and Utku Evci. The dormant neuron phenomenon in deep reinforcement learning. In *Proceedings of the 40th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 2023.
- Sebastian Thrun and Anton Schwartz. Issues in using function approximation for reinforcement learning. In *Proceedings of the Fourth Connectionist Models Summer School*, 1993.
- Mark Towers, Ariel Kwiatkowski, Jordan K. Terry, John U. Balis, Gianluca De Cola, Tristan Deleu, Manuel Goulao, Andreas Kallinteris, Markus Krimmel, Arjun KG, Rodrigo Perez-Vicente, Andrea Pierré, Sander Schulhoff, Jun Jet Tai, Hannah Tze, and Omar G. Younis. Gymnasium: A standard interface for reinforcement learning environments. In *arXiv preprint arXiv:2407.17032*, 2024.
- Hado van Hasselt. Double q-learning. In *Advances in Neural Information Processing Systems*, volume 23, 2010.
- Hado van Hasselt, Arthur Guez, Matteo Hessel, Volodymyr Mnih, and David Silver. Learning values across many orders of magnitude. In *Advances in Neural Information Processing Systems*, volume 29, 2016a.
- Hado van Hasselt, Arthur Guez, and David Silver. Deep reinforcement learning with double Q-Learning. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence*, 2016b.
- Hado van Hasselt, Yotam Doron, Florian Strub, Matteo Hessel, Nicolas Sonnerat, and Joseph Modayil. Deep reinforcement learning and the deadly triad. In *arXiv preprint arXiv:1812.02648*, 2018.
- Shangdong Zhang and Richard S. Sutton. A deeper look at experience replay. In *arXiv preprint arXiv:1712.01275*, 2017.